LEONARDO CHIMAL

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PROFILE

Engineering student with experience in CAD-driven prototyping, robotics, mechanical design, and software development. Skilled in combining mechanical systems, embedded electronics, and software to create functional prototypes. Seeking an engineering internship where I can contribute to product design, prototyping, and real-world system development.

EDUCATION

Black Hawk College — **Moline | Illinois**
GPA | 3.36 — May 2027

United Township High School — **East Moline | Illinois**
GPA | 4.1 (weighted) — May 2025

PROJECTS / SUMMERY / SKILL

Autonomous AI Drone Platform
https://leonardochimal-portfolio.vercel.app/projects/drone

Designed and prototyped a modular 5-inch autonomous drone platform in Onshape integrating embedded electronics, power distribution, vibration-isolated mounting systems, and future autonomous system integration. Focused on iterative CAD development, component integration, weight optimization, and maintainable mechanical design workflows.

Languages: JavaScript, Python, SQL
CAD/Engineering: Onshape, CAD, 3D Printing, PETG, Drone Systems, Embedded
Development: React, Next.js, APIs, AWS

Meta Quest Dual Controller Slide-Lock Attachment
https://leonardochimal-portfolio.vercel.app/projects/meta-quest

Designed and prototyped a custom Meta Quest dual-controller slide-lock attachment focused on improving two-handed stability, ergonomic handling, and mechanical locking functionality through iterative CAD development and physical testing. Developed multiple design iterations using Onshape and PETG 3D printing workflows to refine usability, structural durability, and controller alignment accuracy.

Languages: JavaScript, Python, SQL
CAD/Engineering: Onshape, CAD, 3D Printing, PETG
Prototyping: Mechanical Design, Product Design, Ergonomics

EXPERIENCE

John Deere — **Moline | Illinois**
Apprentice IT — June 2024 – Current

- Built internal tools and automation workflows using AWS, Terraform, JavaScript, and API integrations to support scalable technical systems
- Designed and deployed a secure URL shortener with metadata tracking, encode/decode logic, and backend performance improvements for internal use
- Collaborated with cross-functional teams through GitHub and Agile workflows to troubleshoot issues, document changes, and improve system reliability
- Gained experience working in a professional engineering environment focused on technical problem-solving, documentation, and production-ready solutions

First Tech Challenge — **Team 8813**
Robotics Design Lead — May 2022 – 2025

- Led CAD design and mechanical integration for competitive FTC robotics systems using Onshape, collaborating with programming and manufacturing teams to optimize robot performance, maintainability, and weight distribution through iterative prototyping and engineering reviews.
- Designed drivetrain, intake, launcher, and structural robot systems while working alongside engineers and mentors from John Deere, Lockheed Martin, Meta, Tesla, Amazon, Black Hawk College, and other STEM organizations.
- Helped implement Agile/Scrum-style workflows using Trello, Discord task tracking, and sprint reviews while contributing to STEM outreach, mentoring, engineering presentations, and robotics demonstrations throughout the Quad Cities region.
- Qualified for the FTC World Championship in Houston, Texas (2024), FTC State Championship competitions (2025), and the FTC World Championship in Indianapolis, Indiana (2026).